EEN1095 Git Repository and Development Record

Student Name	Adarsh Pramod	**Student ID**	A12944
Project Title	Prediction of flood using digital twin visualization	**Supervisor**	Dr, Ali Intizar
Implementation Period	Week 1 - Week 8	**Version Date**	02/07/2026
Link to the Repository	https://github.com/adarshpramod2-hub/Flood---Prediction-Digital-Twin	**Shared to Supervisor**	No

Development Record:

Weeks 1-2	
Main implementation / development work completed	The first step of the project was to go through the reference materials of a minimum 25 reference in the initial stage and to see if any changes could be made in the project that is currently being done
Key files, folders, or components added / updated	
Evidence of versioning / commits / iterations	The initial stage involved reviewing academic papers and existing research related to flood prediction and digital twin technology.
Testing, debugging, or validation carried out	
Problems encountered and how they were addressed	
Next development steps	To install the python which shall be working as the back end and the cesiumjs which shall be operated in the front-end

Weeks 3-4	
Main implementation / development work completed	Installation of python and the cesiumJS
Key files, folders, or components added / updated	
Evidence of versioning / commits / iterations	The Backend and the frontend applications (python and cesiumJS) were installed and the parameters which is the backbone of this project has been decided I.e the latitude, longitude, area, and the water depth
Testing, debugging, or validation carried out	
Problems encountered and how they were addressed	
Next development steps	The next development shall be for the raw datas, which shall be found in a few legally authorised sites of the country. A minimum of 500 raw datas are being targeted

Weeks 5-6	
Main implementation / development work completed	To find out the raw datas and see how well it runs.
Key files, folders, or components added / updated	
Evidence of versioning / commits / iterations	The main aim of this weeks work was to findout the datas see how it runs. Currently found few errors which shall be rectifies the next week work report submission
Testing, debugging, or validation carried out	
Problems encountered and how they were addressed	The program was failed and the testing is being done rigourasly

Next development steps	To check if the prediction can be made by using predictive datas rather than the available datas.
Weeks 7-8	
Main implementation / development work completed	Implementation of back end coding (Python)
Key files, folders, or components added / updated	Live data Ingestion- To establish a direct link with the raw data to fetch the real time hydrometric sensor metrics Data fault filtering- Detect broken gauges using error handling block Decoupled Output- Converts datas to JSON file seperating the back end math from the front end map.
Evidence of versioning / commits / iterations	The main aim of this weeks work was to bring in the fully developed backend (python) coding .
Testing, debugging, or validation carried out	A successful testing of the code has been carried out multiple times facing few error, which has been corrected
Problems encountered and how they were addressed	1. Package name error 2. Trailing directory 404 error 3. Loop processing latency 4. Server Bot Block
Next development steps	Developing the cesiumJS and the 3D model digital twin, working on the front end.

Weeks 9-10	
Main implementation / development work completed	
Key files, folders, or components added / updated	
Evidence of versioning / commits / iterations	Enter a short summary of commit activity here. Add commit-history screenshots if applicable.

Testing, debugging, or validation carried out	
Problems encountered and how they were addressed	
Next development steps	Outline the planned development tasks for the next two-week period.
Weeks 11-12	
Main implementation / development work completed	
Key files, folders, or components added / updated	
Evidence of versioning / commits / iterations	Enter a short summary of commit activity here. Add commit-history screenshots if applicable.
Testing, debugging, or validation carried out	
Problems encountered and how they were addressed	
Next development steps	Outline the planned development tasks for the next two-week period.

Weeks 13-14	
Main implementation / development work completed	
Key files, folders, or components added / updated	
Evidence of versioning / commits / iterations	Enter a short summary of commit activity here. Add commit-history screenshots if applicable.

Testing, debugging, or validation carried out	
Problems encountered and how they were addressed	
Next development steps	Outline the planned development tasks for the next two-week period.